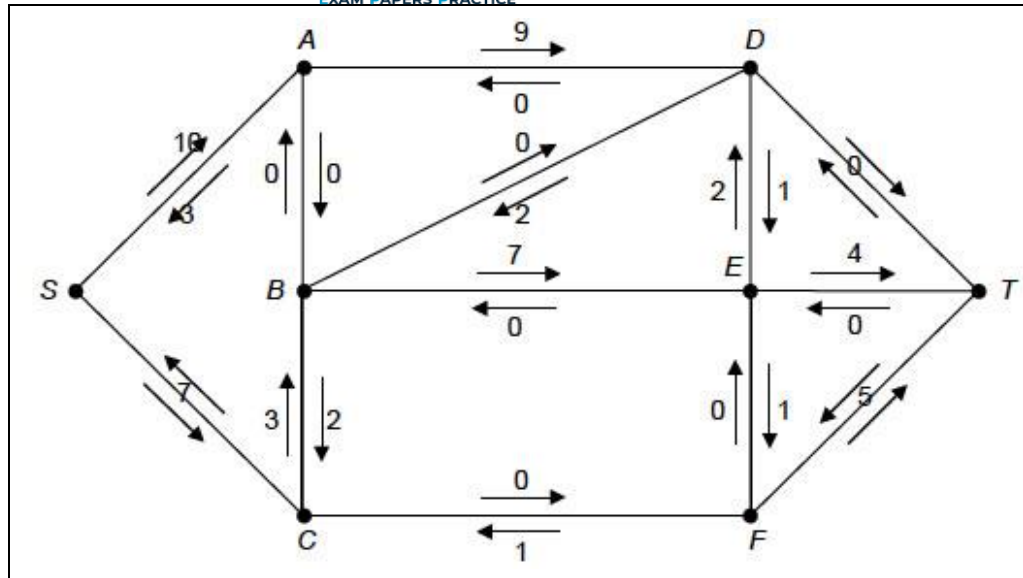


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Explains correctly using maximum flow into and minimum flow out of A	AO2.4	E1	Max flow into vertex $A = 7$ Min flow out of vertex $A = 4 + 3 = 7$ As flow into $A =$ flow out of A , both AD and AB must be at the lower capacity
(i)	Explains correctly using maximum flow into and minimum flow out of E	AO2.4	E1	$AD = 4$ from (a)(i), so $DE = 1$ and $DT = 3$. Since flow out of vertex E is at least 2, and $DE = 1$, BE must be at its upper capacity of 1.
(b)	Explains correctly the statement using a cut in the network	AO2.4	E1	Min flow across the cut $\{S, A, B, C\} \setminus \{D, E, G, T\}$ $= AD + BE + \min(BG) + \min(CG)$ $= 4 + 1 + 5 + 2$ $= 12 > 11$

(c)	Finds a potential increase and decrease for each arc with a value on the two arrows for each arc, with SA, SC, AB, BC and CF correct.	AO1.1a	M1	See diagram below
(i)	Determines correctly the potential increase and decrease for each arc with values on all arrows correct.	AO1.1b	A1	



(ii)	Finds correctly one augmenting path and flow	AO1.1b	B1	<table><tr><th>Augmenting Path</th><th>Flow</th></tr><tr><td>SADEFT</td><td>1</td></tr><tr><td>SCBET</td><td>3</td></tr><tr><td>SADBET</td><td>1</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	Augmenting Path	Flow	SADEFT	1	SCBET	3	SADBET	1				
	Augmenting Path	Flow														
	SADEFT	1														
SCBET	3															
SADBET	1															
	Finds correctly two augmenting paths and the flows	AO1.1b	B1													
	Finds correctly three (or four) augmenting paths and the flows, and clearly states the maximum flow in context with units	AO3.2a	B1	Maximum flow through the network is 22 litres per second												
(iii)	Constructs a rigorous mathematical proof using the value of a cut and the maximum flow-minimum cut theorem (to achieve this mark, the student must clearly show the cut they are calculating the value of and clearly state that the value of this cut is equal to the value of the flow in (c)(ii) and then conclude that the flow is maximal by the maximum flow-minimum cut theorem)	AO2.1	R1	$\{S, A, B, C, D, E\} / \{F, T\} = 7 + 3 + 9 + 3 = 22$ As the flow (22) is equal to the value of the cut (22), the maximum flow is 22 by the maximum flow-minimum cut theorem.												

(iv)	Argues that, as 7 litres per second are initially flowing through node E, only a further 2 litres per second can flow through node E	AO2.4	R1	Flow through network can only increase by 2 litres per second as DT and CF are already saturated Therefore restricted capacity node reduces the maximum flow to $17 + 2 = 19$ litres per second
------	--	-------	----	--



Interprets the impact of the restricted capacity node, concluding that the maximum flow is reduced to 19 litres per second	AO3.A	R1	
Total 11 marks			

Q2.

- (a) (i) Max Flow = 50
(Min cut = 50)

Either statement

E1

- (ii) $35 \leq \text{max flow} \leq 50$
(or min cut)

E1 for strict inequalities

E1, E1

- (iii) Error or contradiction
oe

E1

4

- (b) At F ,

$$\left. \begin{array}{l} \text{flow in} \geq 8 \\ \text{flow out} \leq 7 \end{array} \right\}$$

Stating F and one of the 'flows'

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M1

A1

2

[6]**Q3.**

- (a) $10 + 13 - 1 + 17$

3 values added and -1 (condone one slip)

M1

$$= 39$$

A1

2

- (b) (i) DE 12

on Figure 1

B1

FG 7

B1

2

(ii)

arc	forward	backward
SA	3	1
AB	1	1
BT	0	1
SC	0	2
CA	0	1
AD	0	1
CD	1	1
DE	1	2
BE	1	3
ET	2	3
SF	1	1
FC	1	2
FD	1	0
FG	1	1
DG	0	1
EG	2	1
GT	1	3
	2	

at least 6 pairs correct on Figure 2 (must have arrows)

M1

all correct

A1

2

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(iii) Table

Path	Extra Flow
SABET	1
SFDGT	1
SACDGT	1

1 correct path and extra flow

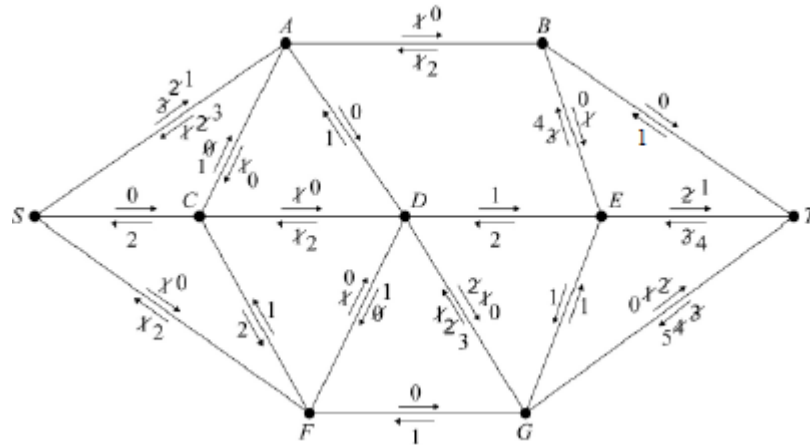
M1

all correct

DEG triangle may have different flows with implications to triangle GET.

A1

Network



1 path correctly augmented forward and backward
but must have earned M1 in part (b)(ii)

network correct

M1

A1

4

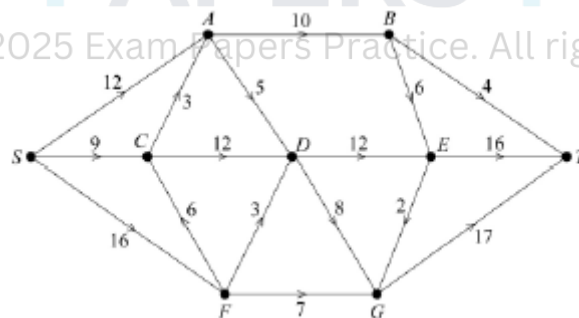
(c) (i) Max flow = 37

B1

1

(ii) Max flow
correct flow of 37

B2



condone 2 slips or omissions in flow of 37 **or** "correct"
feasible flow of 36
for SC1

2

(d) Cut through AB, AD, CD, FD and FG
 $\{S, A, C, F\} \{B, D, E, G, T\}$

B1

1

[14]

Q4.

- (a) Value of cut = $10 + 10 + 15 - 4 - 1$
condone one slip if working shown

M1

= 30

A1

2

- (b) *BT 2, DE 3, ET 12*
any 2 correct

B1

all correct

B1

2

- (c) (i) Initial flows forward and back or double
 Arc with arrows (at **least 6 pairs correct**)
*Condone pairs of values, (coordinates)
 with single arrow*

M1

*all correct (condone pairs with single
 arrow provided key indicated)*

A1

2

(ii)

Path	Flow
SABT	2
SCDET	1
SACBT	1

- (or SCBT instead of SACBT with flow 1
first correct path and flow

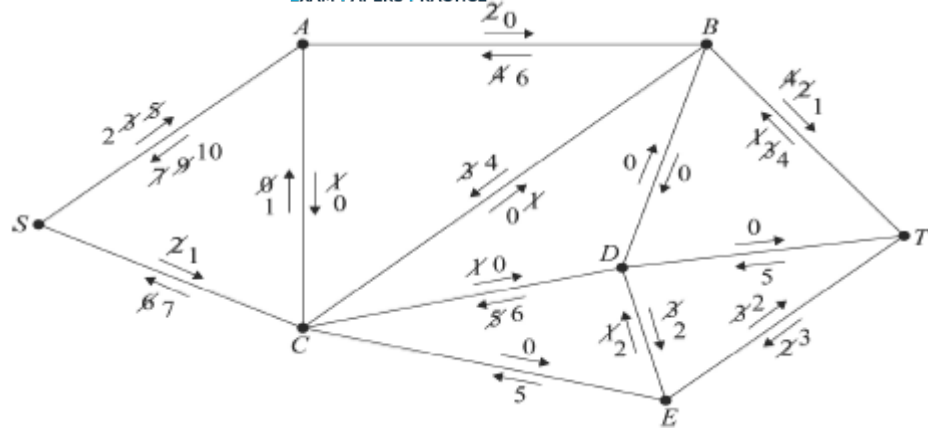
M1

another correct path and flow

A1

*all correct
 (other possibilities also)*

A1



Must have forward and backward flows
augmenting flows (6 pairs correct)

M1

correct

A1

Alternative SA (3 & 9) SC (0&8)

5

(iii)



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May have
SA (14), SC (14)
and AC (4) using
alternative
Maximum flow values

at least 8 correctly interpreted from their
Figure 3 but $24 < \text{their maxflow} < 29$

M1A1

2

- (d) Cut through AB, CB, CD and CE
May use $\{S, A, C\} / \{B, D, E, T\}$

But must have total flow of 28 in their
network (condone one slip)

B1B1

1

[14]

Q5.

- (a) Value of cut = $30 - 10 + 12 + 20$

M1

= 52

Full marks for correct answers without working

A1

2

- (b) $AE = 9$;

B1

$EF = 5$;

B1

$FG = 4$

B1

3

- (c) (i) Attempt at forward and backward flows

$SA \ 2 \ \& \ 4$; $AB \ 1 \ \& \ 3$; $BT \ 1 \ \& \ 3$

$SD \ 3 \ \& \ 1$; $DA \ 0 \ \& \ 3$; $AE \ 0 \ \& \ 3$

$BE \ 0 \ \& \ 7$; $DE \ 2 \ \& \ 0$; $ET \ 1 \ \& \ 3$

$FD \ 2 \ \& \ 1$; $EF \ 5 \ \& \ 1$; $EG \ 1 \ \& \ 5$

$FG \ 1 \ \& \ 2$; $GT \ 3 \ \& \ 0$

At least 5 pairs correct

M1

10 pairs correct

A1

all correct

A1

3

- (ii) First flow augmenting path and correct flow on table

M1

Table correct

A1

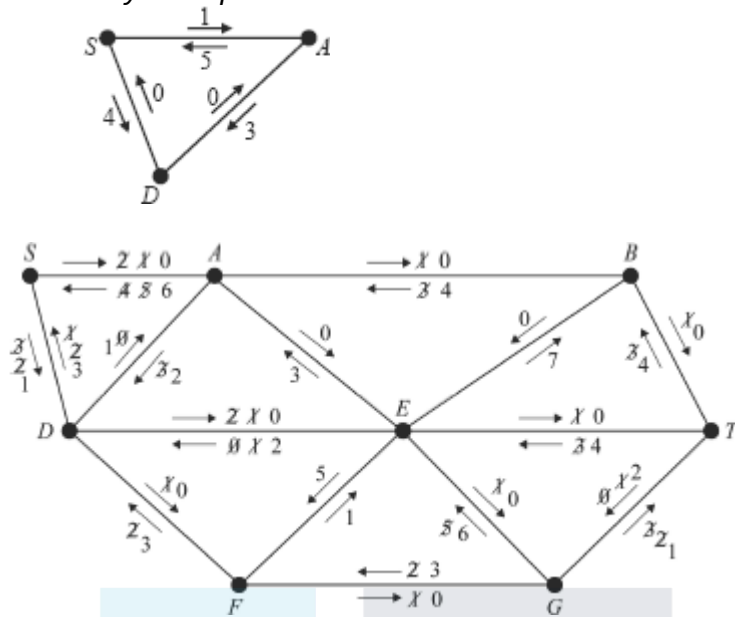
Adjusting flows – forward and back

M1

Correct

A1

May end up with



Path	Extra flow	
SABT	1	Or SDET
SADET	1	
SDFGT	1	
SDEGT	1	

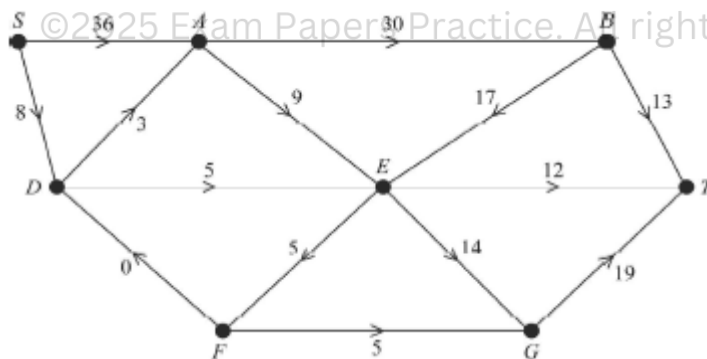
- (d) Max flow of 44 shown on figure 3
up to 2 slips

M1

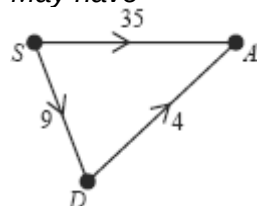
all correct

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A1



May have



(e) Cut through their saturated arcs

M1

Cut passes through AB, AE, DE and DF
 Or BT, ET, EG , and FG

A1

2

[16]

Q6.

(a) (i) $17 - 9 + 16 + 20 = 44$

B1

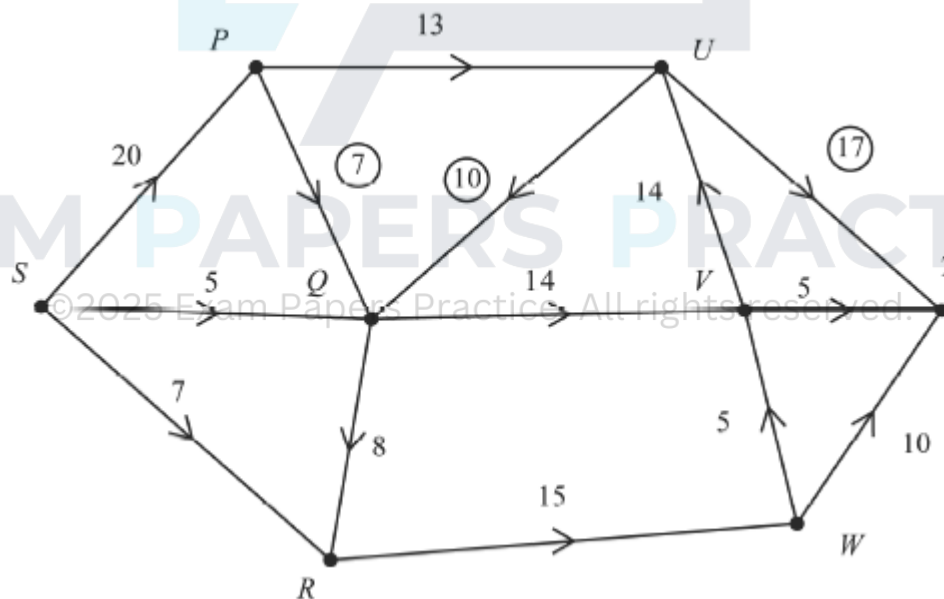
1

(ii) Max flow ≤ 44

B1ft

1

(b)



7

B1

10

B1

17

B1

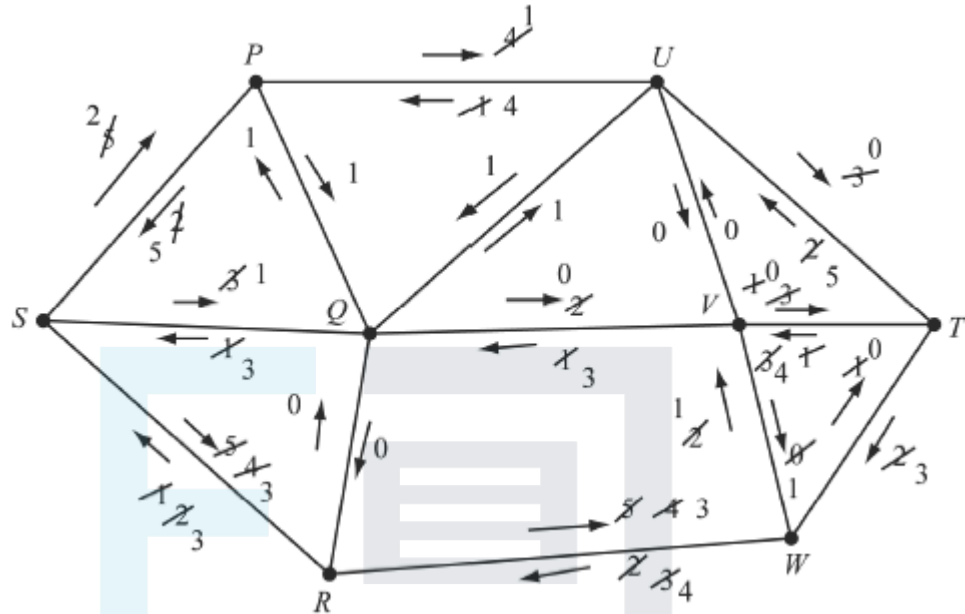
3

- (c) (i) Initial forward and backward flows
5 pairs correct

M1

Correct

A1



2

(ii)

Path	Additional Flow
SPUT	3
SQVT	2
SRWT	1
SRWVT	1

adjusting flows on network (1 path shown correctly)

M1

correct

A1

additional flow in table

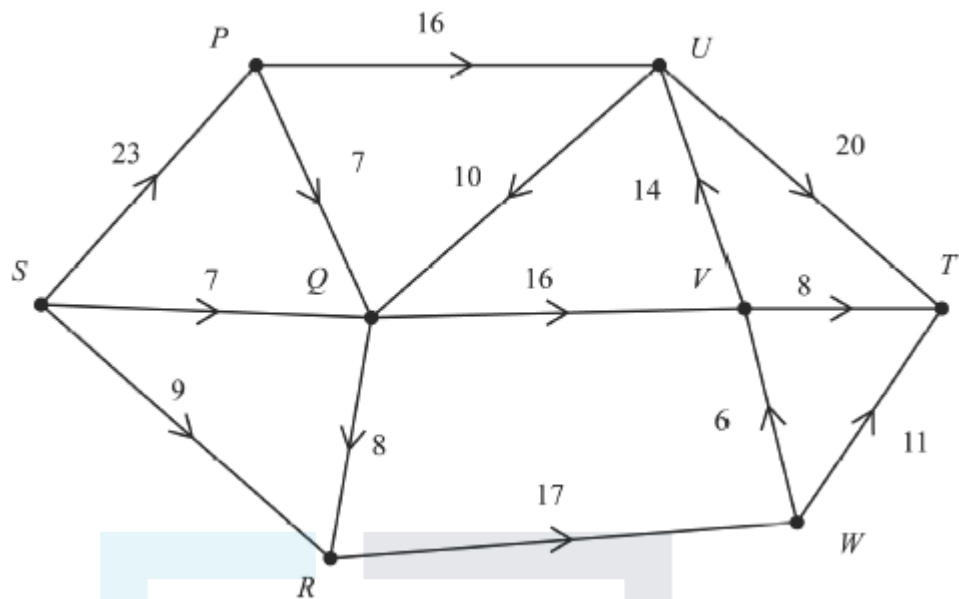
M1

second flow

A1

all correct

(iii)



Max flow of 39 (several possibilities of final flow diagram)

B1

1

[13]

Q7.

(a) (i) $5 + 8 + 16 - 3 = 26$

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B1

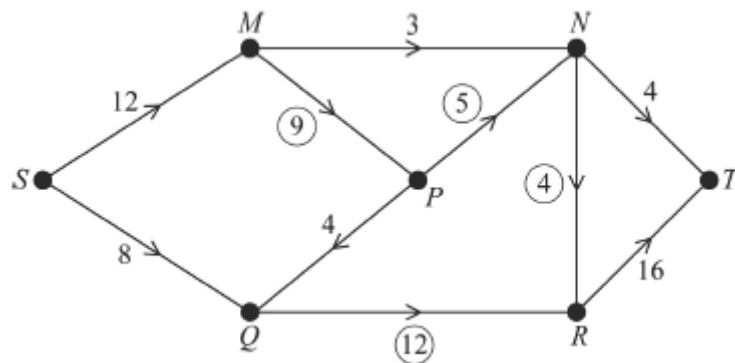
1

(ii) Max flow ≤ 26

E1ft

1

(b)



MP - 9

B1

PN – 5

B1

NR – 4

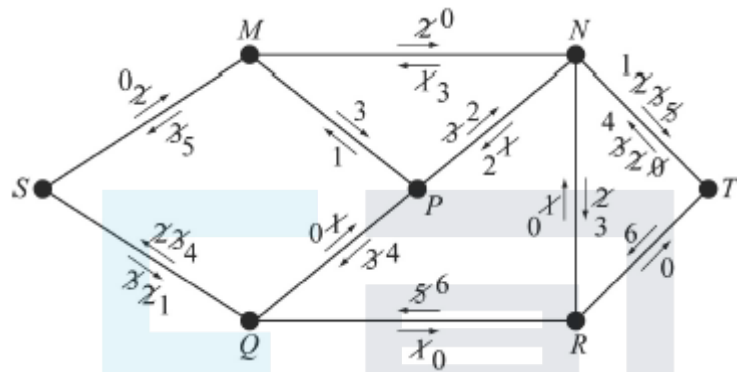
B1

QR – 12

B1

4

(c) (i)



initial flow – forward and backward
6 pairs correct

M1

correct

OM 2 & 3; MN 2 & 1
NT 5 & 0; MP 3 & 1
SQ 3 & 2; PQ 3 & 1
PN 3 & 1; QR 1 & 5
NR 2 & 1; RT 0 & 6

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A1

2

(ii) Adjusting flows on network

M1A1

Path	Flow
SMNT	2
SQPNT	1
SQRNT	1

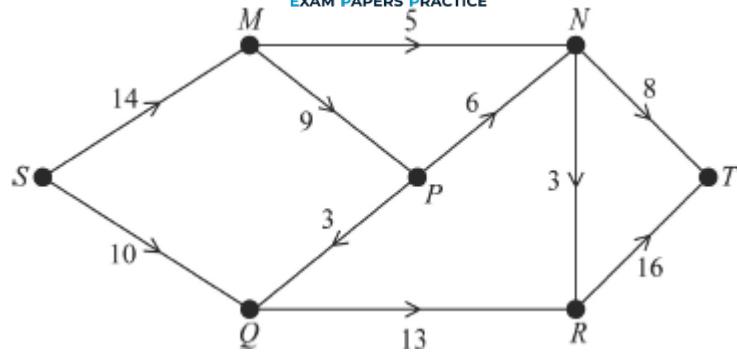
First correct path and flow B1

Second correct B1

Rest B1

5

(iii)

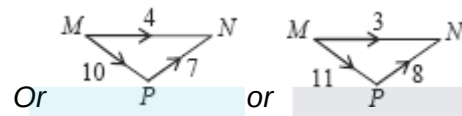


6 flows correct

all correct

M1

A1

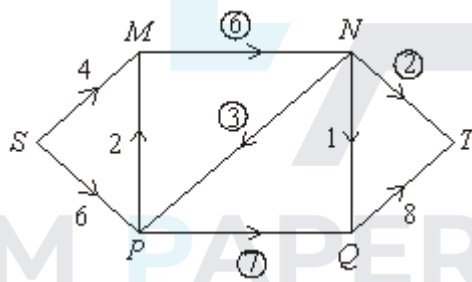


2

[15]

Q8.

(a)



MN

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B1

NT

B1

PQ

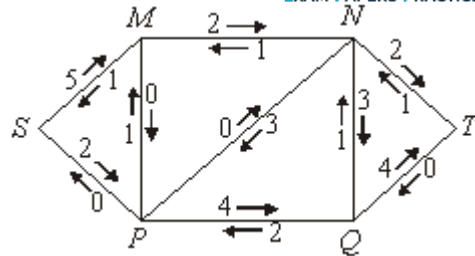
B1

NP

B1

4

(b) (i)



initial flow indicated as surplus forward and backward flows

M1

use of flow augmentation

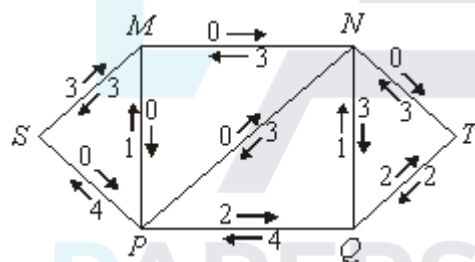
M1

e.g SMNT 2
one flow correctly identified

A1

SPQT 2
all possible flows correct

A1



amending flows (dep on first M1)

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M1

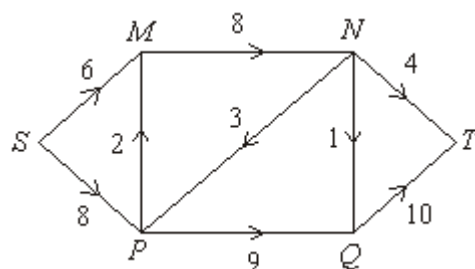
final situation with saturation
at M and P

A1

6

(ii) Max flow – 14

B1



B1
2

- (c) Cut through 2 of their saturated arcs
cut on **original** network

M1

described or drawn

A1
2

[14]

Q9.

- (a) (i) Flow = 25

E1

- (ii) $15 \leq \text{Flow} \leq 25$

E1
2

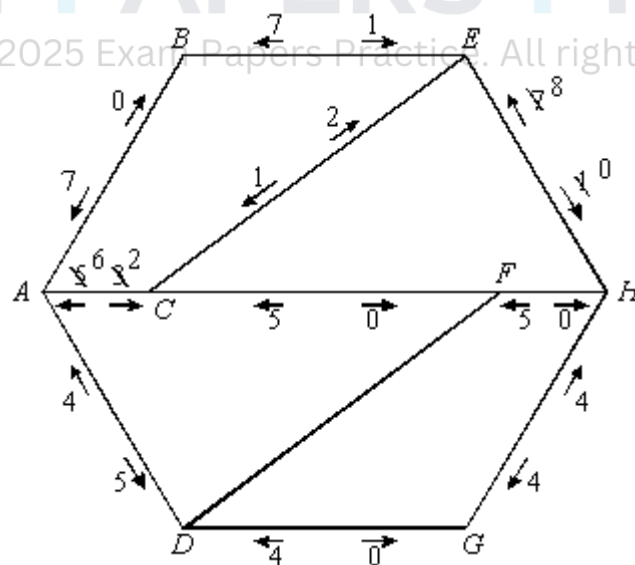
- (b) At F
Min in = 7

M1

Max out = 5

A1
2

- (c) (i)



SCA

M1

1 complete flow

M1 A1

2nd complete flow

A1

All correct

A1

5

(ii) Any cut = 17

B1

1

[10]



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Examiner reports

Q2.

This question proved to be a challenge for students. Students were required to think of the implications of statements and not merely apply an algorithm. Part (a)(i) was answered quite well, but in part (a)(ii), many students failed to state a range for the maximum flow. Some students realised that there had to be an error in part (a)(iii). Fully correct answers to part (b) were rarely seen. Many students highlighted the problem at a vertex other than 'F' and justified their answer by considering maximum flow in and minimum flow out.

Q3.

Part (a) It was surprising to see a large number of able candidates being unable to calculate the correct value of the cut, with many confusing upper and lower capacities in their calculation.

Part (b)(i) Few candidates had trouble in finding the flows along DE and FG .

Part (b)(ii) It was pleasing to see that many more candidates this year were aware of how to show the potential increases and decreases along each edge. Those who did not do this lost both marks here and two marks in the next part of the question.

Part (b)(iii) It was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. These candidates used the table effectively to show what new flows had been introduced and modified both the forward and backward flows on their network. Marks were awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation; candidates are advised to show the initial flow in ink and then to augment their flow using pencil.

Part (c) Those who stated that the maximum flow was 36 were given some credit for illustrating a flow of 36 on Figure 3. Those with the correct maximum flow were usually able to show a correct flow on the network; those who made a couple of slips were again given some credit for their work.

Part (d) Many candidates were able to find the correct cut with capacity 37, but quite a few who had the correct maximum flow were deceived into using their Figure 3 on which every cut had value 37, rather than looking at the initial network with upper and lower capacities.

Q4.

Part (a) Most candidates seemed unaware of how to calculate the value of the cut correctly because of the upper and lower capacities. Some subtracted 9 instead of 5 from the sum of the other upper capacities; others merely added a set of positive numbers.

Part (b) In contrast, almost everyone scored full marks for finding the value of each of the missing flows along the given edges.

Part (c)(i) Some candidates still seemed unaware of how to represent the potential increases and decreases **from the initial feasible flow**. It requires forward and backward arrows or a duplicate edge; one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows so that any adjustments can be shown in pencil so as not to obliterate the initial flow figures.

Part (c)(ii) A table was provided so that a flow of 2 along *SABT*, for example, could be listed in the table. The potential forward and backward flows along *SA*, *AB* and *BT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. If candidates obliterate their values of the potential increases and decreases from part(i) then they risk losing marks for that part. The majority of candidates found only 2 augmenting paths and were not able to find the correct maximum flow. Those who were not using the labelling procedure correctly sometimes obtained a flow that was impossible; for instance a maximum flow of 29 was seen quite often. It should be noted that candidates are not expected to do flow augmentation with negative flows.

Part (c)(iii) Some credit was given for transferring values from Figure 1 to Figure 2 even if the maximum flow had not been achieved; but full marks could only be scored for a correct flow of 28 through the network.

Part (d) Very few candidates realised the need to consider their saturated arcs in order to obtain the correct cut. Many wrongly considered their flow on Figure 2, where obviously every cut has a value of 28, and consequently wrote down cuts such as the one through *BT*, *DT* and *ET* as their final answer. That is why the question suggested that candidates “may wish to show the cut on the network above” since that network contained all upper and lower capacities.

Q5.

In part (a), most candidates seemed unaware of how to calculate the value of the cut correctly. Many subtracted 17 instead of 10 from the sum of the other upper capacities.

In contrast, in part (b), almost everyone scored full marks for finding the value of each of the missing flows along the given edges.

Some candidates still seemed unaware, in part (c)(i), of how to represent the potential increases and decreases **from the initial feasible flow**. It requires forward and backward arrows or a duplicate edge: one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows so that any adjustments can best be shown in pencil so as not to obliterate the initial flow figures. In part (c)(ii), a table was provided so that a flow of 1 along *SABT*, for example, could be listed in the table. The potential forward and backward flows along *SA*, *AB* and *BT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. If candidates obliterate their values of the potential increases and decreases from part (c)(i), then they risk losing marks for that part. The majority of candidates found only 3 augmenting paths and were not able to find the correct maximum flow.

In part (d), those who used flow augmentation correctly usually had no trouble in completing the diagram to show a maximum flow of 44 litres per second.

In part (e), very few candidates realised the need to consider their saturated arcs in order to obtain a minimum cut. Many wrongly considered their flow on Figure 3, where obviously every cut has a value of 44, and consequently wrote down cuts such as *BT*, *ET* and *GT* as their final answer.

Q6.

In part (a), about half the candidates seemed unaware of how to calculate the value of the cut correctly. In part (b), by contrast, almost everyone scored full marks for finding the value of the missing flows along the given edges.

In part (c)(i), some candidates still seemed unaware of how to represent the potential increases and decreases **from the initial feasible flow**. It required forward and backward arrows or a duplicate edge: one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows, with any adjustments then shown in pencil so as not to obliterate the initial flow figures.

A table was provided in part (c)(ii) so that a flow of 3 along *SPUT*, for example, could be listed in the table. The potential forward and backward flows along *SP*, *PU* and *UT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. There are many correct solutions to this part, and teachers might find it informative to consider different possibilities with their future students in order to illustrate that, whatever solution they obtain, the maximum flow is still the same.

In part (c)(iii), those who used flow augmentation correctly usually had no trouble in completing the diagram to show a maximum flow of 39 litres per second.

Q7.

In part (a), very few candidates seemed prepared for a question with lower capacities as well as upper capacities, and consequently the correct value of the cut was rarely seen.

In contrast, in part (b), almost everyone scored full marks for the missing flows along the given edges.

In part (c)(i), many candidates seemed unaware of how to represent the potential increases and decreases from the initial flow. It requires forward and backward arrows or a duplicate edge: one showing the potential forward flow, the other the potential backward flow. In part (c)(ii), flow augmentation was not widely understood. A table was provided so that a flow of 2 along *SMNT* could be listed in the table. The potential forward and backward flows along *SM*, *MN* and *NT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. Candidates who used flow augmentation correctly usually had no trouble in completing the diagram in part (c)(iii) to show a maximum flow of 24.

Q8.

The insert seemed to help candidates to set out their solution in a logical manner. Most candidates completed the first network indicating the feasible flow. When doing the flow augmentation in part (b)(i), candidates rarely showed their working properly. It is in cases like this, with minimum and maximum capacities, that the formal method (i.e. two arrows and numbers for each edge showing potential increases and decreases of capacity) can be very helpful for candidates and they should be encouraged to always use it. Teachers might again find it valuable to consult the solution in the mark scheme and to encourage their students to take this approach. Also, in trying to find a cut of 14, only a minority of candidates used the original capacities in the network printed in the question paper.

Q9.

Candidates' answers given in part (a) were not very accurate, nor were the descriptions given for no possible flow in part (b). In part (b), candidates merely had to refer to the restrictions at *F*. The final part of the question was well answered but a small number of candidates simply gave a final flow without any indication of method.